

HTTP Log Analysis

Splunk Basics – HTTP Log Analysis Lab Report

STRUCTURED LAB REPORT

OBJECTIVE

The objective of this lab is to learn how **HTTP web traffic logs can be analyzed using Splunk** in order to detect web-based attacks and suspicious activity.

During this lab we will:

- Upload **Zeek-style HTTP logs** into Splunk.
- Analyze **web traffic generated by clients**.
- Detect **server errors and suspicious user agents**.
- Identify **large file downloads or uploads**.

HTTP log analysis helps security teams detect:

- Web scanning tools
- Automated attack scripts
- Data exfiltration through large transfers
- Suspicious web access patterns

TOOLS AND ENVIRONMENT

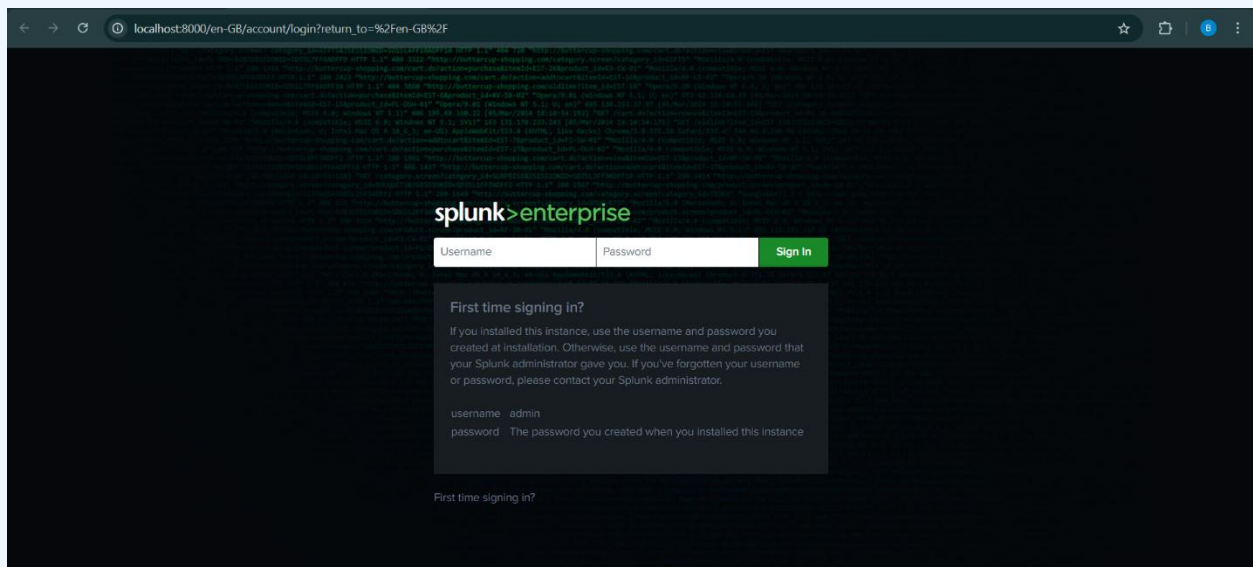
Component	Description
Splunk	Log monitoring and data analysis platform
Data Source	Zeek HTTP logs
Log Format	JSON
Index Used	http_lab
Sourcetype	json

LAB SETUP

Before starting the analysis, the HTTP log file must be uploaded into Splunk.

Step 1 – Open Splunk Web

1. Open **Splunk Web Interface** in your browser.
2. Log in using your credentials.

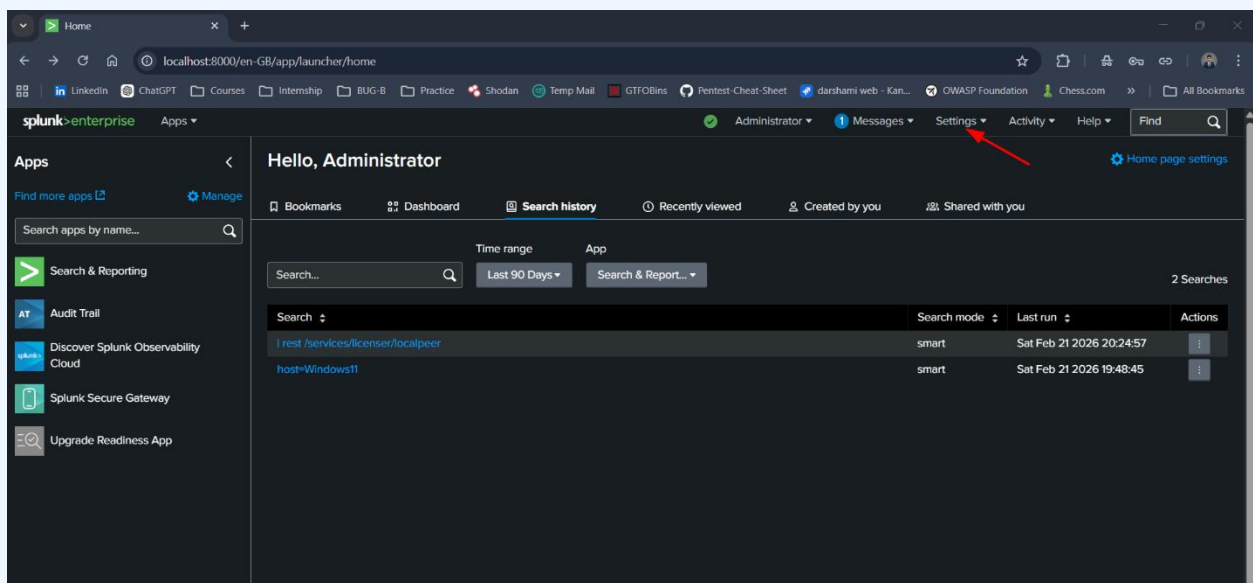


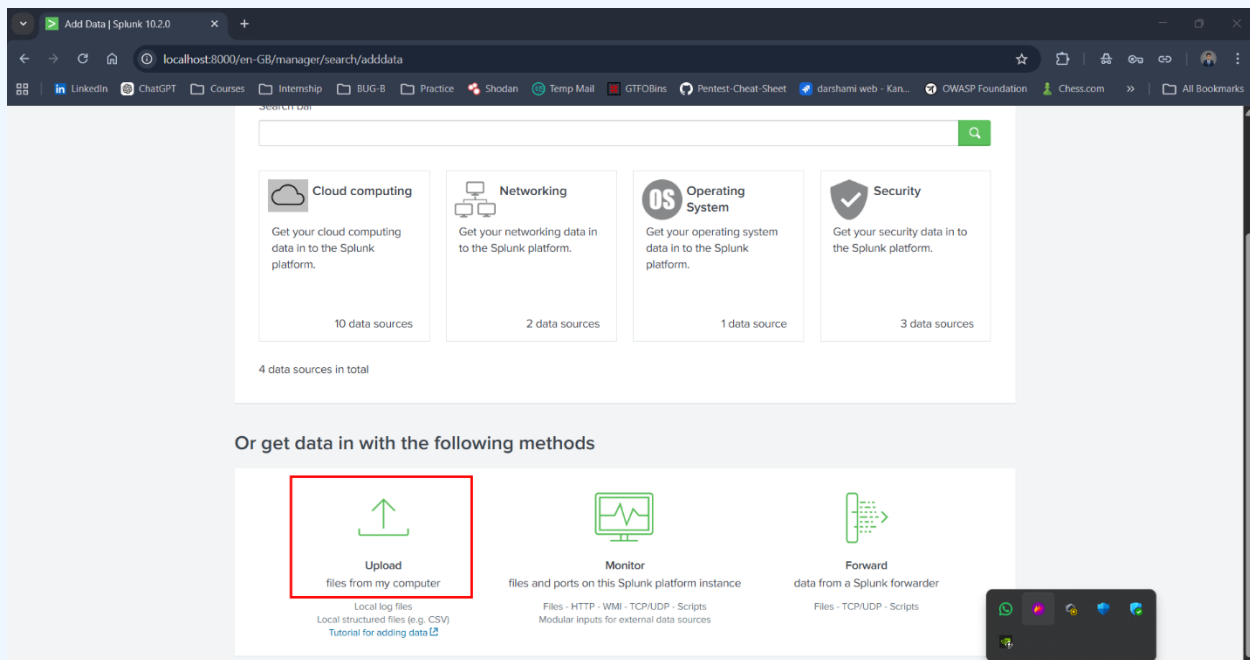
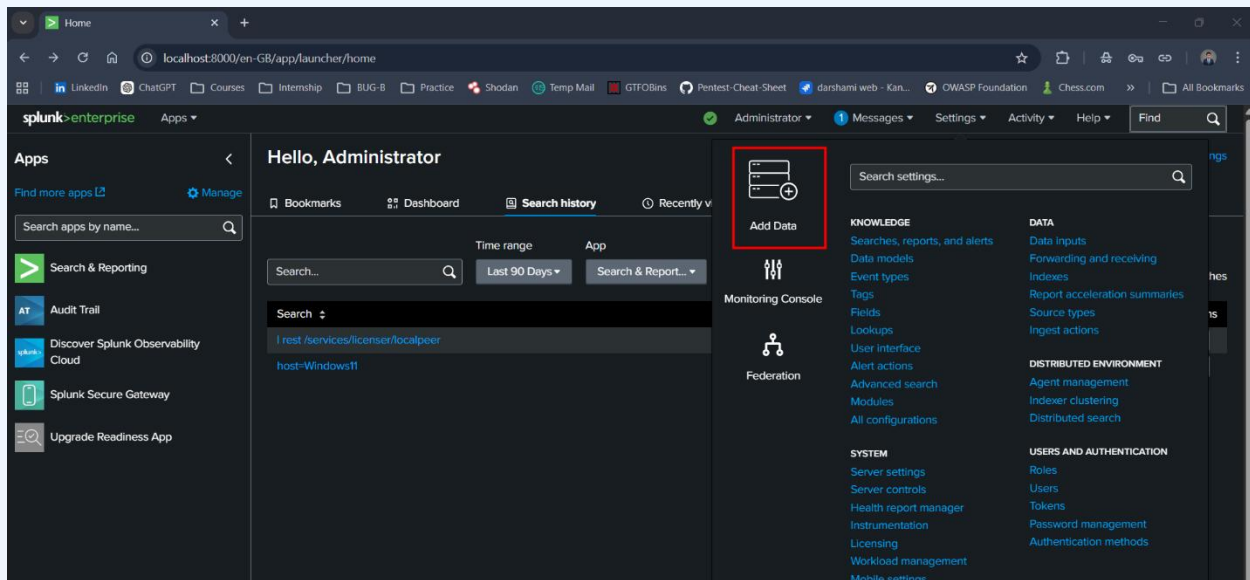
Step 2 – Navigate to Add Data

1. Go to:

Settings → Add Data

2. Select **Upload**.



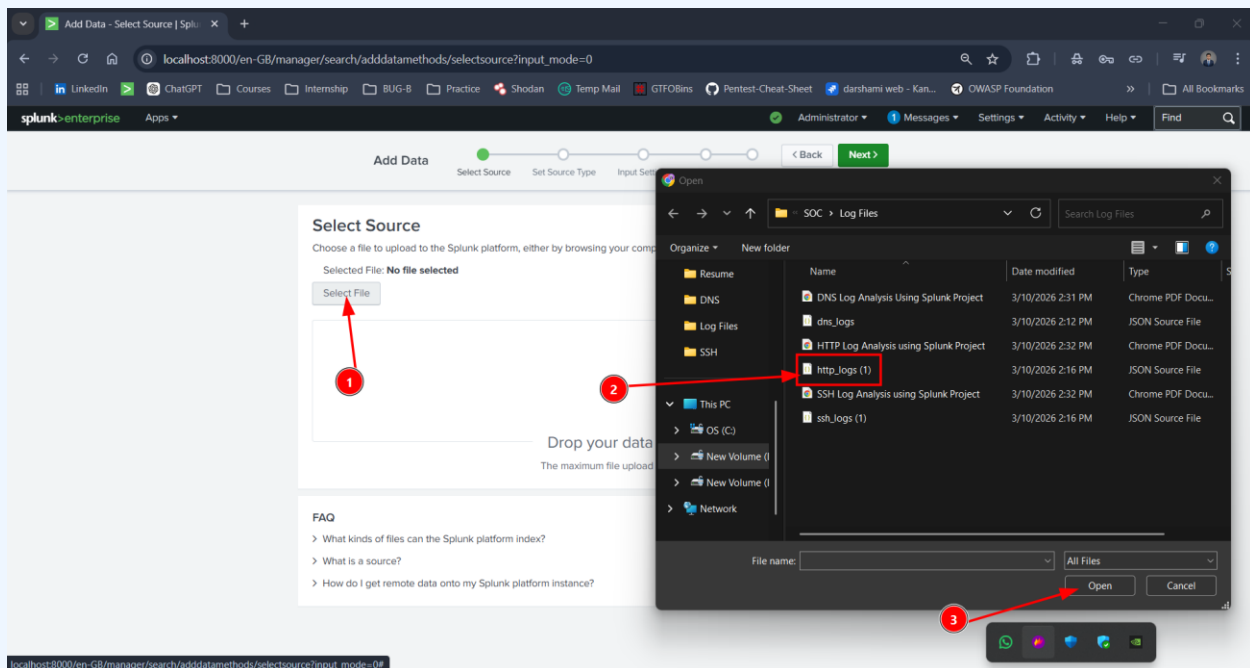


Step 3 – Upload the HTTP Log File

1. Click **Select File**
2. Upload the file:

`http_logs.json`

This dataset contains **HTTP requests and responses recorded by Zeek network monitoring tool.**



Step 4 – Configure Source Type

Set the source type to:

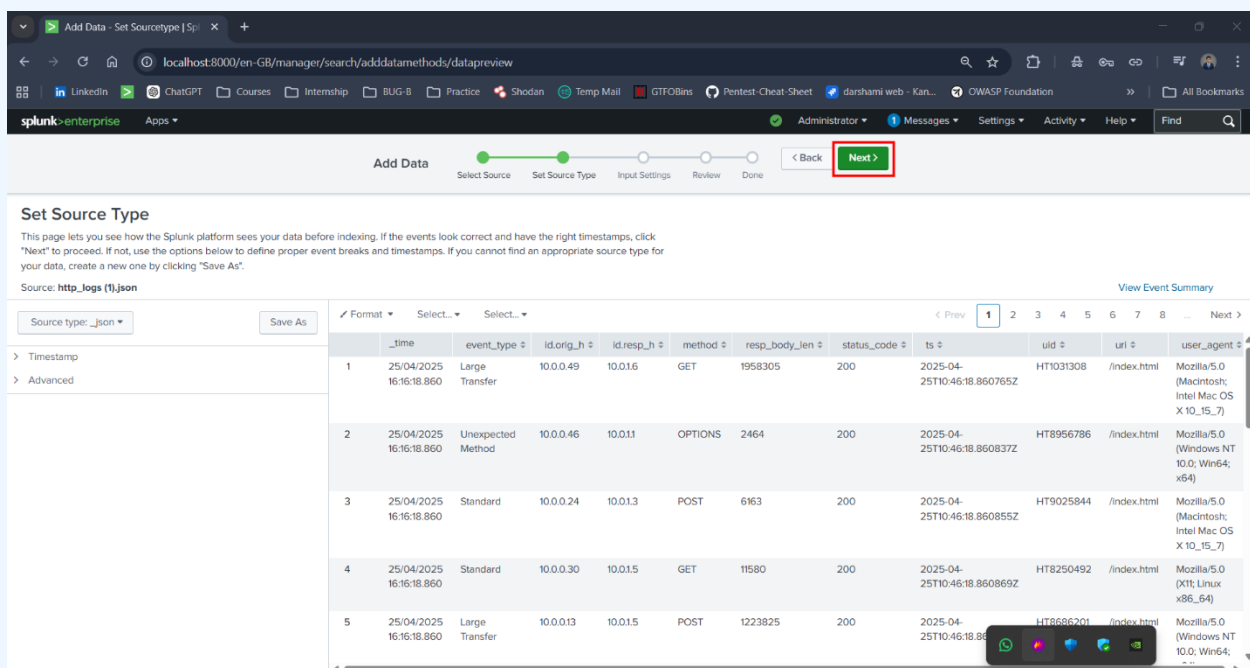
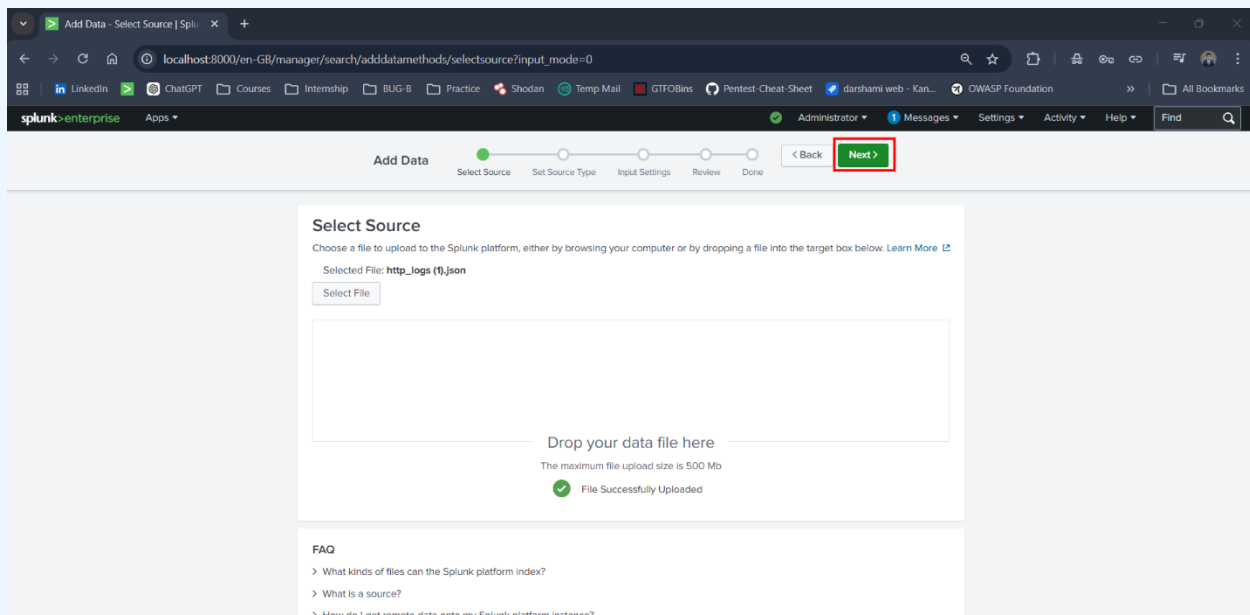
`_json`

or create a custom source type:

`http_logs`

Reason:

- The logs are structured in **JSON format**
- Splunk automatically extracts important fields like:
 - `id.orig_h`
 - `id.resp_h`
 - `uri`
 - `user_agent`
 - `status_code`
 - `resp_body_len`



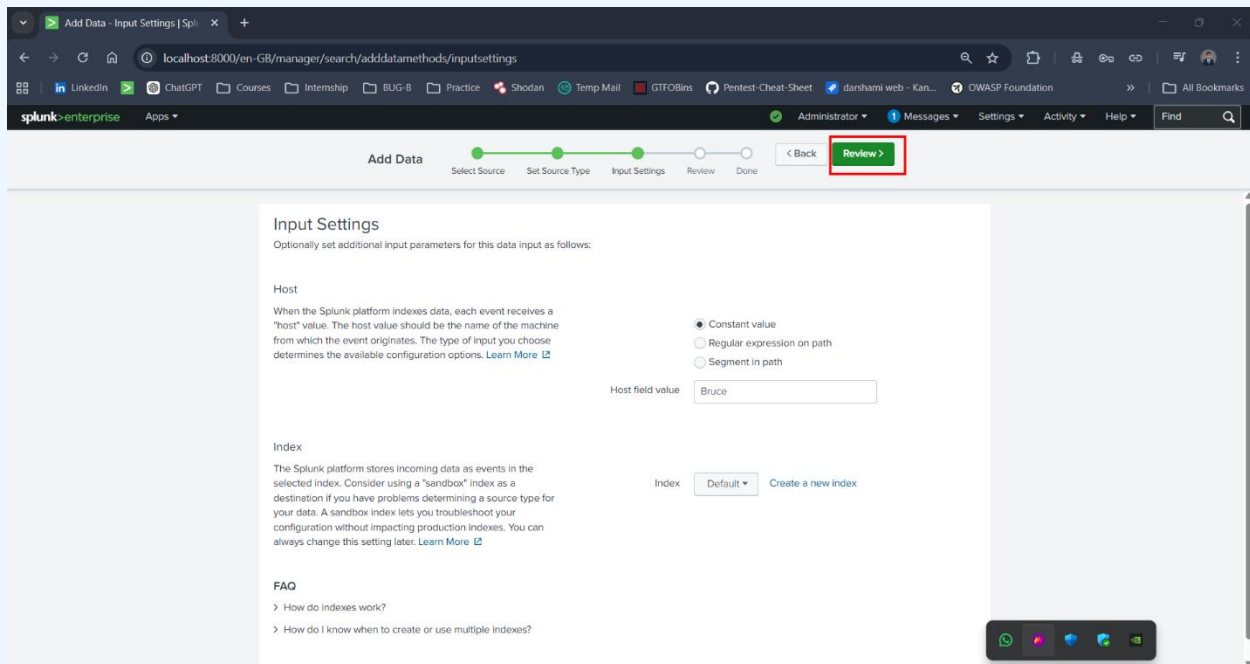
Step 5 – Configure Index

Select or create the index:

http_lab

Reason:

- Keeps HTTP analysis data separate from other logs.
- Makes searching faster and easier.



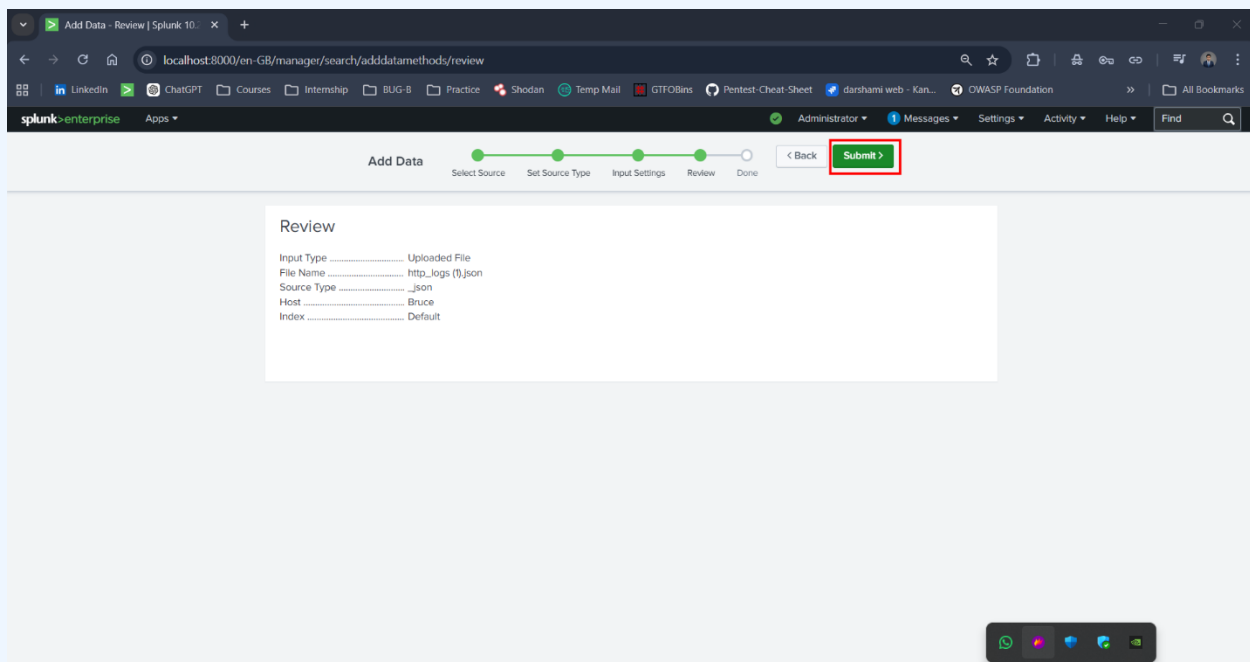
Step 6 – Confirm Upload

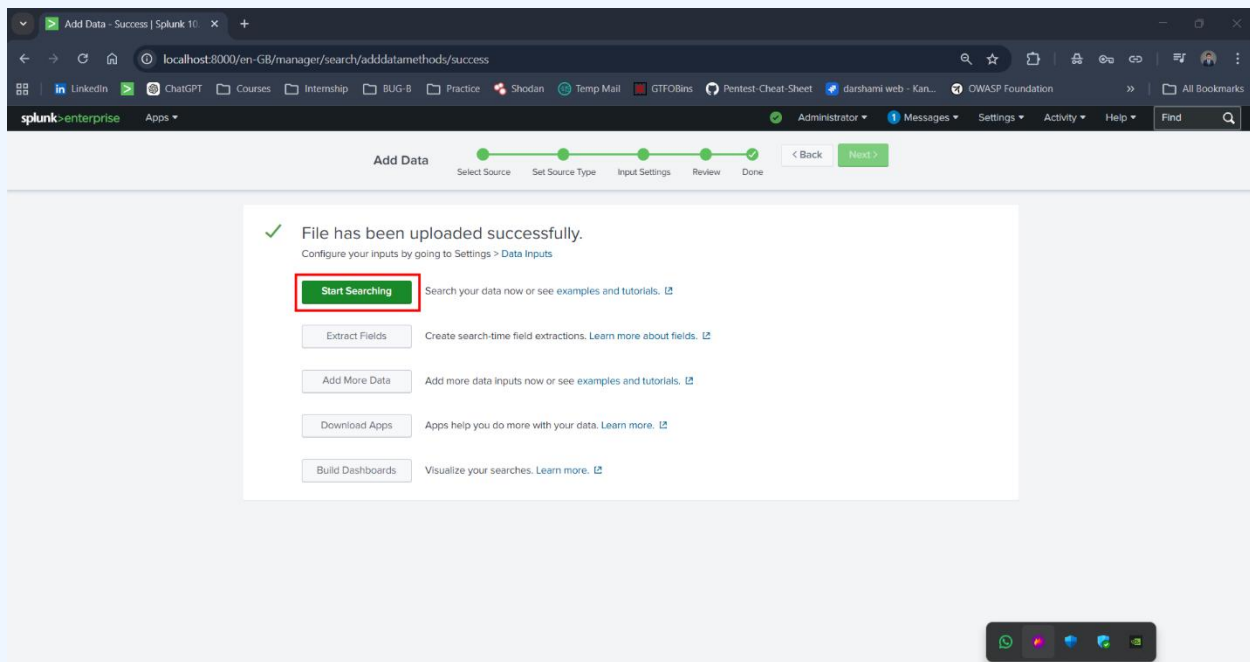
1. Review the configuration.
2. Click **Submit**.

To confirm indexing run:

```
index=http_lab
```

If events appear, the data ingestion was successful.





UNDERSTANDING HTTP LOGS

HTTP logs record **web requests and responses between clients and servers.**

Common fields in Zeek HTTP logs:

Field	Description
id.orig_h	Client IP address
id.resp_h	Web server IP
uri	Requested resource
user_agent	Client application making request
status_code	HTTP response status
resp_body_len	Size of server response

Example HTTP Log Event:

```
{
  "id.orig_h": "192.168.1.10",
  "id.resp_h": "192.168.1.5",
  "uri": "/login.php",
  "user_agent": "curl/7.68.0",
  "status_code": 200,
  "resp_body_len": 10240
}
```

The screenshot shows the Splunk Enterprise interface. The search bar at the top contains the query: `source=http_logs (1).json host=Bruce sourcetype=json`. The search results show 3,000 events. The first event is expanded, showing details such as `event_type: Standard`, `id.orig_h: 10.0.0.17`, `id.resp_h: 10.0.1.10`, `method: POST`, `resp_body_len: 5699`, `status_code: 200`, `ts: 2025-04-25T10:46:18.894102Z`, `uid: Hf4913534`, `uri: /index.html`, and `user_agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)`. Red circles 1, 2, and 3 highlight the search bar, the search button, and the expanded event details respectively.

LAB TASKS

Task 1: Find Top 10 Endpoints Generating Web Traffic

SPL Query

```
index=http_lab sourcetype="json"
| stats count by "id.orig_h"
| sort -count
| head 10
```

Explanation

Command	Purpose
<code>index=http_lab</code>	Searches logs stored in http_lab
<code>sourcetype="json"</code>	Filters JSON logs
<code>stats count by id.orig_h</code>	Counts HTTP requests from each client IP
<code>sort -count</code>	Sorts results in descending order
<code>head 10</code>	Displays the top 10 IP addresses

What This Shows

This query helps identify:

- Clients generating the most web traffic
- Potential scanning hosts
- Systems performing automated requests

The screenshot shows the Splunk Search interface. The search bar contains the query: `source=http_logs (1).json host="Bruce" sourcetype="json" | stats count by "id.orig_h" | sort -count | head 10`. The results are displayed in a table with two columns: `id.orig_h` and `count`. The top 10 results are shown, sorted by count in descending order. Red boxes and arrows highlight the query, the search button, and the result columns.

id.orig_h	count
10.0.0.28	76
10.0.0.31	73
10.0.0.42	73
10.0.0.27	72
10.0.0.48	70
10.0.0.45	70
10.0.0.14	69
10.0.0.58	67
10.0.0.13	65
10.0.0.25	65

Task 2: Count Server Errors (HTTP 5xx)

HTTP **5xx errors** indicate problems on the server side.

SPL Query

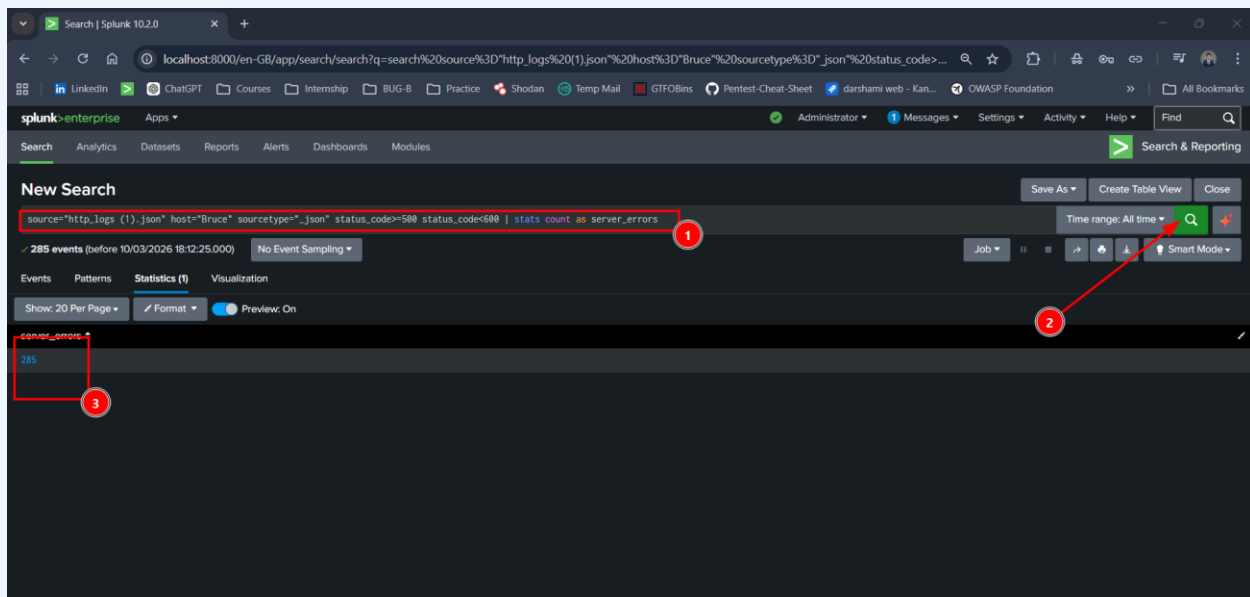
```
index=http_lab sourcetype="json" status_code>=500 status_code<600  
| stats count as server_errors
```

Explanation

Command	Purpose
<code>status_code>=500 status_code<600</code>	Filters server error responses
<code>stats count</code>	Counts matching events
<code>as server_errors</code>	Renames result field

What This Shows

This query identifies **how many server-side errors occurred**.



Large numbers of 5xx errors may indicate:

- Web application issues
- Server overload
- Exploitation attempts

Task 3: Identify User Agents Associated with Scripted Attacks

Attack tools often use specific **user-agent strings**.

SPL Query

```
index=http_lab sourcetype="json" user_agent IN ("sqlmap/1.5.1",  
"curl/7.68.0", "python-requests/2.25.1", "botnet-checker/1.0")  
| stats count by user_agent
```

Explanation

Command	Purpose
<code>user_agent IN (...)</code>	Filters requests from known attack tools
<code>stats count by user_agent</code>	Counts occurrences per user agent

Known Suspicious Tools

User Agent	Description
<code>sqlmap</code>	SQL injection testing tool
<code>curl</code>	Command-line HTTP client
<code>python-requests</code>	Python automation library
<code>botnet-checker</code>	Possible botnet testing tool

The screenshot shows the Splunk Search interface. A search query is entered in the 'New Search' bar: `source="http_logs (1).json" host="Bruce" sourcetype="json" user_agent IN ("sqlmap/1.5.1", "curl/7.68.0", "python-requests/2.25.1", "botnet-checker/1.0") | stats count by user_agent`. The search results are displayed in a table view, showing the user agent and the count of occurrences. The table has two columns: 'user_agent' and 'count'. The results are as follows:

user_agent	count
botnet-checker/1.0	78
curl/7.68.0	69
python-requests/2.25.1	78
sqlmap/1.5.1	79

Red annotations are present: a red box around the search query (1), a red arrow pointing to the search button (2), and a red box around the search results table (3).

These tools may indicate **automated scanning or exploitation attempts**.

Task 4: Identify Large File Transfers (>500 KB)

Large transfers may indicate **file downloads, uploads, or data exfiltration**.

SPL Query

```
index=http_lab sourcetype="json" resp_body_len>500000
| table ts "id.orig_h" "id.resp_h" uri resp_body_len
| sort -resp_body_len
```

Explanation

Command	Purpose
<code>resp_body_len>500000</code>	Filters responses larger than 500 KB
<code>table</code>	Displays selected fields
<code>sort -resp_body_len</code>	Sorts results by largest transfer

What This Shows

This query identifies:

- Large downloads
- Possible data exfiltration
- File transfers over HTTP

The screenshot shows the Splunk Enterprise interface. The search bar contains the query: `source="http_logs (*)" host="Bruce" sourcetype="json" resp_body_len>500000 | table ts "id.orig_h" "id.resp_h" url resp_body_len | sort -resp_body_len`. The search results show 323 events. The table displays columns: `ts`, `id.orig_h`, `id.resp_h`, `url`, and `resp_body_len`. The results are sorted by `resp_body_len` in descending order. The first few rows show large response body lengths, indicating potential data exfiltration or large file transfers.

ts	id.orig_h	id.resp_h	url	resp_body_len
2025-04-25T10:46:18.879523Z	10.0.0.23	10.0.1.4	/index.html	1977613
2025-04-25T10:46:18.875312Z	10.0.0.49	10.0.1.7	/index.html	1976613
2025-04-25T10:46:18.888207Z	10.0.0.50	10.0.1.3	/index.html	1974922
2025-04-25T10:46:18.863055Z	10.0.0.30	10.0.1.5	/index.html	1968142
2025-04-25T10:46:18.861070Z	10.0.0.48	10.0.1.12	/index.html	1960325
2025-04-25T10:46:18.860765Z	10.0.0.49	10.0.1.6	/index.html	1958305
2025-04-25T10:46:18.862785Z	10.0.0.57	10.0.1.10	/index.html	1951520
2025-04-25T10:46:18.882673Z	10.0.0.36	10.0.1.7	/index.html	1945121
2025-04-25T10:46:18.877884Z	10.0.0.15	10.0.1.10	/index.html	1944872
2025-04-25T10:46:18.880465Z	10.0.0.35	10.0.1.2	/index.html	1941685
2025-04-25T10:46:18.874011Z	10.0.0.43	10.0.1.11	/index.html	1929703
2025-04-25T10:46:18.882569Z	10.0.0.28	10.0.1.5	/index.html	1924376
2025-04-25T10:46:18.887168Z	10.0.0.32	10.0.1.4	/index.html	1916823
2025-04-25T10:46:18.883728Z	10.0.0.51	10.0.1.9	/index.html	1909326

Large transfers could indicate **file downloads or data leakage**.

SECURITY INSIGHTS

HTTP log analysis can reveal many types of attacks:

1. **Web Scanning:** Attack tools sending many automated requests.
2. **SQL Injection Testing:** Tools like **sqlmap** probing web applications.
3. **Botnet Communication:** Suspicious user agents connecting to servers.
4. **Data Exfiltration:** Large file transfers leaving the network.

Splunk enables **real-time monitoring and detection of these activities**.

CONCLUSION

In this lab we successfully:

- Uploaded **Zeek HTTP logs into Splunk**
- Learned how Splunk processes **JSON formatted web logs**
- Used **SPL queries to analyze HTTP traffic**

The analysis helped detect:

- Top clients generating web traffic
- Server-side errors
- Suspicious automated tools
- Large file transfers

These techniques are important for **web security monitoring, intrusion detection, and incident response**.